

EC2 Backend Deployment

by Dinusha Ekanayake — PredictiX Project

Instance Details

Item	Value
Provider	AWS EC2
Instance type	t3.micro (free tier eligible)
OS	Ubuntu 26.04 "Resolute"
Region	ap-south-1 (Mumbai)
Public IP	13.232.200.154
Backend URL	https://13.232.200.154.sslip.io
Repo path on EC2	/home/ubuntu/PredictiX_Backend
SSH user	ubuntu
Key file	C:\Users\Dinusha Ekanayake\Downloads\predictix-key.pem

1. AWS Setup

Key Pair

Type: RSA, .pem format. Downloaded to Windows as **predictix-key.pem**.

Fixed Windows permissions (CodexSandboxUsers group caused "Bad permissions" SSH error):

```
$keyPath = "C:\Users\Dinusha Ekanayake\Downloads\predictix-key.pem"
$acl = Get-Acl $keyPath
$acl.SetAccessRuleProtection($true, $false)
$acl.Access | ForEach-Object { $acl.RemoveAccessRule($_) }
$rule = New-Object System.Security.AccessControl.FileSystemAccessRule($env:USERNAME, "Read", "Allow")
$acl.AddAccessRule($rule)
Set-Acl $keyPath $acl
```

Security Group Inbound Rules

Port	Protocol	Source	Purpose
22	TCP	0.0.0.0/0	SSH
80	TCP	0.0.0.0/0	HTTP (Certbot ACME challenge)
443	TCP	0.0.0.0/0	HTTPS
8000	TCP	0.0.0.0/0	FastAPI direct (optional)

2. Server Setup

SSH Connection

```
ssh -i "C:\Users\Dinusha Ekanayake\Downloads\predictix-key.pem" ubuntu@13.232.200.154
```

System Packages

Ubuntu 26.04 ships Python 3.14 — python3.12 does not exist on this OS.

```
sudo apt update && sudo apt upgrade -y
sudo apt install -y python3.14 python3.14-venv python3-pip git nginx certbot python3-certbot-nginx
```

3. Code Deployment

Clone Repository

```
cd /home/ubuntu
git clone https://YOUR_GITHUB_TOKEN@github.com/Dinusha-Ekanayake/PredictiX_Backend.git
cd PredictiX_Backend
git checkout dev
```

Python Virtual Environment

```
python3.14 -m venv venv
source venv/bin/activate
```

requirements-prod.txt (created on EC2)

Excluded: torch, transformers, sentence-transformers, huggingface_hub, safetensors — these exceed the 8 GB EBS disk. HuggingFace models run remotely via HF Hub (DISABLE_HF_MODELS=true).

```
fastapi uvicorn pandas numpy scikit-learn catboost shap xgboost
lifelines joblib groq nltk langchain-groq langchain-core reportlab
sqlalchemy alembic apscheduler psycpg[binary] psycpg2-binary
python-jose passlib[bcrypt]==1.7.4 bcrypt==4.1.2 pydantic-settings
python-multipart requests python-dotenv email-validator supabase

pip install -r requirements-prod.txt
pip install xgboost # installed separately after initial disk space issue
```

.env File (/home/ubuntu/PredictiX_Backend/.env)

```
DATABASE_URL=...
SUPABASE_URL=...
SUPABASE_KEY=...
HF_TOKEN=...
HF_ASSET_SUMMARIZATION_REPO=...
GROQ_API_KEY=...
DISABLE_HF_MODELS=true
ALLOWED_ORIGINS=https://13.232.200.154.sslip.io
```

4. ML Model Files

All .pkl files in the git repo are **Git LFS pointers** (134-byte text files) — not real binaries. Real model files were copied from Windows to EC2 using scp.

Files Copied via scp

File	Size	Destination
predictix_xgboost_classifier_v6.pkl	5.7 MB	pdm_classifier_model/

maintenance_classifier_features.pkl	573 B	pdm_classifier_model/
predictix_pm_model_v5.pkl	97 MB	pdm_regressor_model/
regression_selected_features.pkl	558 B	pdm_regressor_model/
battery_aft.pkl	~858 KB	survival_analysis/
brake_aft.pkl	~858 KB	survival_analysis/
hydraulic_aft.pkl	~858 KB	survival_analysis/
oil_aft.pkl	~858 KB	survival_analysis/
tire_aft.pkl	~858 KB	survival_analysis/

scp Command Format (from Windows PowerShell)

```
scp -i "C:\Users\Dinusha Ekanayake\Downloads\predictix-key.pem" `
    "e:\...\model.pkl" `
    ubuntu@13.232.200.154:/home/ubuntu/PredictIX_Backend/app/ai/models/pdm_regressor_model/
```

WARNING: Never git commit these files. Back up to /tmp/ before any git reset --hard, then restore after. /tmp/ is wiped on instance reboot.

5. Code Fixes Applied

Fix 1: ModuleNotFoundError: transformers at startup

File: app/ai/services/asset_summary_service.py

Moved 'from transformers import ...' inside the function, guarded by DISABLE_HF_MODELS check (lazy import).

Fix 2: ModuleNotFoundError: sentence_transformers at startup

File: app/ai/services/knowledge_service.py

Moved 'from sentence_transformers import SentenceTransformer' inside _get_model() (lazy import).

Fix 3: invalid load key, 'v' — LFS pointer instead of real pkl

Copied real binary files via scp.

Fix 4: No module named 'predictor_name' — wrong loader

File: app/main.py — _load_pickle()

Added joblib fallback (regressor bundle was saved with joblib.dump, not pickle):

```
def _load_pickle(path: Path):
    try:
        with open(path, 'rb') as fh:
            return pickle.load(fh)
    except Exception:
        return joblib.load(path)
```

Fix 5: AttributeError: 'dict' object has no attribute 'predict'

File: app/main.py — _load_pdm_models()

Both pkl files are dict bundles. Unwrap them on load: Classifier: bundle['model'] (XGBoost), bundle['threshold'], bundle['feature_cols'], bundle['categorical_cols'] Regressor: bundle['explainer_model'] (CatBoost — use this), bundle['feature_cols'], bundle['categorical_cols']

```

clf_bundle = _load_pickle(CLF_MODEL_PATH)
if isinstance(clf_bundle, dict):
    clf_model          = clf_bundle['model']
    clf_features        = list(clf_bundle.get('feature_cols', []))
    clf_threshold       = float(clf_bundle.get('threshold', 0.5))
    clf_categorical_cols = list(clf_bundle.get('categorical_cols', []))

reg_bundle = _load_pickle(REG_MODEL_PATH)
if isinstance(reg_bundle, dict):
    reg_model          = reg_bundle['explainer_model'] # CatBoost
    reg_features        = list(reg_bundle.get('feature_cols', []))
    reg_categorical_cols = list(reg_bundle.get('categorical_cols', []))

```

Also updated **app/ai/services/batch_prediction_service.py**: `_run_regressor()`, `run_batch_for_asset()`, and `run_batch_for_all_assets()` each gained a `reg_categorical_cols` parameter (removed `get_cat_feature_indices()` call).

6. systemd Service

File: `/etc/systemd/system/predictix.service`

```

[Unit]
Description=PredictiX FastAPI Backend
After=network.target

[Service]
User=ubuntu
WorkingDirectory=/home/ubuntu/PredictiX_Backend
EnvironmentFile=/home/ubuntu/PredictiX_Backend/.env
ExecStart=/home/ubuntu/PredictiX_Backend/venv/bin/uvicorn app.main:app \
    --host 0.0.0.0 --port 8000 --workers 1
Restart=always
RestartSec=5

[Install]
WantedBy=multi-user.target

sudo systemctl daemon-reload
sudo systemctl enable predictix
sudo systemctl start predictix

```

7. Nginx + HTTPS (sslip.io + Certbot)

Why needed: Frontend is on Vercel (HTTPS). Browsers block HTTP API calls from HTTPS pages (Mixed Content error).

sslip.io: Free IP-based DNS. `13.232.200.154.sslip.io` automatically resolves to `13.232.200.154`. No registration required.

Install + Issue Certificate

```

sudo apt install -y nginx certbot python3-certbot-nginx
sudo certbot --nginx -d 13.232.200.154.sslip.io

```

`/etc/nginx/sites-available/default` — HTTPS Block

```

server {
    listen 443 ssl;
    server_name 13.232.200.154.sslip.io;

```

```

ssl_certificate /etc/letsencrypt/live/13.232.200.154.sslip.io/fullchain.pem;
ssl_certificate_key /etc/letsencrypt/live/13.232.200.154.sslip.io/privkey.pem;

location / {
    proxy_pass http://127.0.0.1:8000;
    proxy_set_header Host $host;
    proxy_set_header X-Real-IP $remote_addr;
    proxy_set_header X-Forwarded-For $proxy_add_x_forwarded_for;
    proxy_set_header X-Forwarded-Proto $scheme;
}
}

```

8. Frontend Configuration

Vercel environment variable:

```
NEXT_PUBLIC_API_URL = https://13.232.200.154.sslip.io
```

CORS origins hardcoded in app/main.py:

```

"https://predicti-x-frontend.vercel.app",
"https://predicti-x-frontend-dinusha-ekanayakes-projects.vercel.app",

```

9. Service Management Commands

Task	Command (from SSH session)
Check status	<code>sudo systemctl status predictix --no-pager</code>
View live logs	<code>sudo journalctl -u predictix -f</code>
Start service	<code>sudo systemctl start predictix</code>
Stop service	<code>sudo systemctl stop predictix</code>
Restart service	<code>sudo systemctl restart predictix</code>
Pull latest + restart	<code>git pull origin dev && sudo systemctl restart predictix</code>
Check memory	<code>free -h</code>
Check disk	<code>df -h</code>

SSH from Windows:

```
ssh -i "C:\Users\Dinusha Ekanayake\Downloads\predictix-key.pem" ubuntu@13.232.200.154
```

10. Important Notes

- Binary model files are NOT in git — re-copy via scp if the instance is recreated or after git reset --hard.
- Back up binaries to /tmp/ on EC2 before any reset. /tmp/ is wiped on instance reboot — re-copy from Windows if needed.
- If the EC2 instance is stopped and started, the public IP may change — update NEXT_PUBLIC_API_URL in Vercel.
- Free tier duration: 12 months from account creation, then ~\$7.50/month for t3.micro.
- Memory usage at idle: ~430 MB — tight on 1 GB RAM. Avoid triggering multiple batch predictions simultaneously.
- DISABLE_HF_MODELS=true — torch/transformers are NOT installed locally. HF inference runs remotely via HF Hub.

